

Unit 1: Ratios and Proportions

Topic 1: Word Problems — Challenge Level 1 🍌🍌

Name: _____

Date: _____

Q1. The ratio of dogs to cats is 4:5. If there are 36 animals in total, how many are dogs?

- (A) 16
- (B) 20
- (C) 9
- (D) 12

Q2. A mixture uses a 2:7 ratio of syrup to water. How much syrup is in 45 liters of the mixture?

- (A) 10L
- (B) 35L
- (C) 14L
- (D) 20L

Q3. Simplify the ratio 1.5 : 4.5.

- (A) 1:3
- (B) 3:9
- (C) 15:45
- (D) 2:6

Q4. Which ratio is equivalent to 8:12 and 10:15?

- (A) 1:2
- (B) 2:3
- (C) 3:4
- (D) 4:5

Q5. In a bag of 50 marbles, the ratio of blue to green is 3:7. How many more green marbles are there than blue?

- (A) 15
- (B) 35
- (C) 20
- (D) 10

Q6. Scale of a drawing is 2cm : 5m. If a wall is 20m long, how long is it in the drawing?

- (A) 4cm
- (B) 8cm
- (C) 10cm
- (D) 50cm

Q7. A tape diagram shows 3 units for Group A and 5 units for Group B. If one unit equals 4, what is the total?

- (A) 12
- (B) 20
- (C) 32
- (D) 8

Q8. [FRQ] Use a tape diagram to solve: The ratio of pens to pencils is 3:2. There are 15 more pens than pencils. How many pencils are there?

Q9. [FRQ] Express the ratio $\frac{1}{2} : \frac{3}{4}$ in simplest whole number form. Show your work using a common denominator.

Q10. [FRQ] A birdhouse has a ratio of wings to beaks of 2:1. If the total number of wings and beaks is 60, how many birds are there?

💡 Chef's Tips

- **Total Parts:** In "total" problems, add the parts of the ratio first (e.g., $4 + 5 = 9$ parts).
- **Difference:** If you know the difference between groups, find out what "1 unit" in your ratio represents.
- **Fractions:** To simplify ratios with fractions, multiply both sides by the Least Common Denominator (LCD).